

Why Mechanical Seals Matter:

The Physics Behind the Thin Film That Keeps Industry Running

A Technical Blog Post for Engineers and Curious Minds

Abstract. Mechanical face seals are among the most consequential components in rotating machinery, yet they are routinely under-appreciated until they fail. A film of process fluid less than two micrometres thick — narrower than a bacterium — simultaneously seals, lubricates, and cools the sealing interface. This document explains how that film forms, why it is governed by a cubic (h^3) leakage law that appears to contradict real-world observations, and why four coupled high-order mechanisms resolve the apparent paradox. The governing equations are derived from first principles and supplemented with worked examples. Key references include Lebeck (1991), Hamrock *et al.* (2004), and the Reynolds (1886) lubrication equation.

Introduction: A Seal the Size of a Hand, a Gap the Width of a Germ

Every pump, compressor, mixer, and hydraulic turbine that exists to move fluid faces a fundamental challenge: the rotating shaft must exit the pressurised casing without letting the process fluid escape. Gland packing — braided rings compressed around the shaft — was the historical answer, but it leaks by design, erodes the shaft, and requires continuous adjustment. The modern mechanical face seal solves this with two flat, precision-lapped rings pressed face-to-face across the shaft penetration. Under rotation, a self-regulating fluid film forms between the faces and maintains a near-zero visible leak rate throughout the machine's life.

What makes this engineering feat so remarkable is scale. The film is typically $0.5\text{--}2\text{ }\mu\text{m}$ thick — smaller than a red blood cell. Yet the forces acting across it are substantial: process pressures up to tens of bar, face rubbing velocities of $5\text{--}25\text{ m/s}$, and thermal gradients that cause the seal rings to deform elastically by amounts that, in absolute terms, are comparable to the film itself. Understanding how stability emerges from this environment is the purpose of this report.

Anatomy of a Mechanical Face Seal

A mechanical face seal has five functional elements, illustrated in cross-section in Figure 1.

1. **Primary seal faces.** One face rotates with the shaft; the other is stationary in the gland (housing). The sealing interface is between these two lapped faces. Typical material pairings: hard face of silicon carbide (SiC) or tungsten carbide against a softer carbon-graphite face. The pairing is chosen for low friction, chemical compatibility, and thermal conductivity.
2. **Secondary seals.** O-rings or bellows that seal the faces to the shaft and housing, preventing leakage around (rather than between) the faces. In a *pusher seal* the secondary O-ring slides dynamically along the shaft to follow face wear and axial shaft movement; in a *non-pusher (bellows) seal* a metal or elastomeric bellows flexes instead, accommodating movement without a sliding interface.
3. **Spring or bellows closing element.** Provides the mechanical closing force that keeps the faces in contact (or near-contact), compensates for face wear, and tracks axial shaft movement up to the seal's designed travel limit.
4. **Drive mechanism.** Pins, drive bands, or set-screws that transmit shaft torque to the

rotating face without over-constraining it.

5. **Gland and hardware.** The stationary plate that carries the secondary seal, flush ports, and the assembly to the housing.

Balanced vs. unbalanced seals. The *balance ratio* B describes what fraction of the sealed pressure area acts to close the faces. An unbalanced seal ($B > 0.8$) has a higher hydraulic closing force and suits low pressure; a balanced seal ($B \approx 0.7\text{--}0.8$) reduces the net closing force and is required at higher pressure/speed to prevent film collapse and thermal overload.

Cartridge vs. component seals. A cartridge seal is factory-assembled on a sleeve and installs as one unit, eliminating field measurement of face setting length — the dominant source of installation error. Split seals are cut into halves so they can be fitted around the shaft without dismantling the machine, a critical advantage on large or hard-to-disassemble equipment.

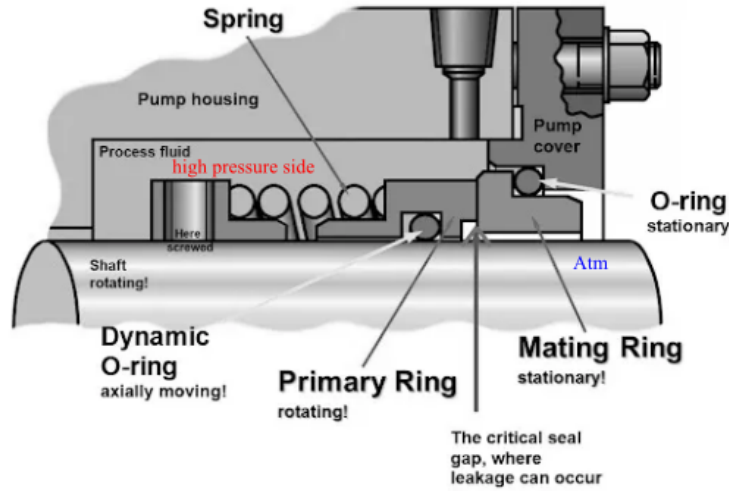


Figure 1: Simplified cross-section of a single pusher-type mechanical face seal. The film between the two primary faces (exaggerated for clarity) is the sealing interface. The dynamic O-ring slides axially along the shaft sleeve to follow wear and shaft movement.

The Two Fundamental Flows: Reynolds' Equation in Cylindrical Coordinates

The entire physics of the seal film is governed by the **Reynolds lubrication equation** (Reynolds, 1886), derived by simplifying the Navier-Stokes equations for a thin viscous film between two surfaces:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(\frac{r h^3}{12\mu} \frac{\partial p}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{h^3}{12\mu} \frac{\partial p}{\partial \theta} \right) = \frac{U}{2} \frac{\partial h}{\partial r} + \frac{\partial h}{\partial t} \quad (1)$$

where r is the radial coordinate, θ the circumferential angle, h the local film thickness, μ the dynamic viscosity, p the film pressure, $U = \omega r$ the local sliding velocity (shaft surface speed), and t time.

The right-hand side encodes how surface motion and squeeze generate pressure; the left-hand side encodes how that pressure drives viscous flow. Two physically distinct flow contributions can be identified:

Couette Flow — Circumferential, Rotation-Driven

The term $\frac{U}{2} \frac{\partial h}{\partial r}$ on the right of Eq. (1) is the **Couette** (shear-driven) source. The rotating face drags the fluid circumferentially at local speed $U = \omega r$. In the gap, the velocity profile varies

linearly from U at the rotating face to 0 at the stationary face, producing a shear stress:

$$\tau = \mu \frac{U}{h} = \frac{\mu \omega r}{h} \quad (2)$$

Couette flow is *circumferential* — it goes around the ring in the direction of shaft rotation. It does **not** transport fluid radially; it carries no fluid from the high-pressure (inner) to the low-pressure (outer) side. Its engineering importance is twofold: (a) it generates frictional heat $q = \tau U = \mu \omega^2 r^2 / h$, which drives thermoelastic deformation of the faces; and (b) wherever the film thickness h varies circumferentially ($\partial h / \partial \theta \neq 0$), the Couette term generates pressure — the hydrodynamic *wedge effect* discussed in Section 5.

Poiseuille Flow — Radial, Pressure-Driven (the Leakage)

The left-hand side of Eq. (1) governs pressure-driven (**Poiseuille**) flow. Across the seal face, the dominant gradient is *radial*: from sealed pressure P_i at the inner radius r_i to atmospheric pressure P_o at the outer radius r_o . For a circumferentially uniform film (steady state, no waviness) this radial Poiseuille flow reduces to:

$$\frac{1}{r} \frac{d}{dr} \left(\frac{r h^3}{12 \mu} \frac{dp}{dr} \right) = 0 \quad (3)$$

Integrating twice with boundary conditions $p(r_i) = P_i$ and $p(r_o) = P_o$ gives the pressure profile across the face:

$$p(r) = P_i + (P_o - P_i) \frac{\ln(r/r_i)}{\ln(r_o/r_i)} \quad (4)$$

and the volumetric leakage rate:

$$Q_{\text{leak}} = \frac{\pi h^3 \Delta P}{6 \mu \ln(r_o/r_i)} \quad (5)$$

where $\Delta P = |P_i - P_o|$ is the sealed pressure differential.

The h^3 scaling is the single most important result in seal engineering. Doubling the film thickness increases leakage by a factor of $2^3 = 8$; halving it reduces leakage by a factor of 8. The seal's job is, fundamentally, to maintain h at the smallest stable value consistent with adequate lubrication and cooling.

The two flows are **orthogonal in direction**: Couette travels *circumferentially*; Poiseuille travels *radially*. They do not interfere with each other, but they are coupled through h and the temperature field, which is why seal analysis requires a fully coupled thermo-fluid-structural model.

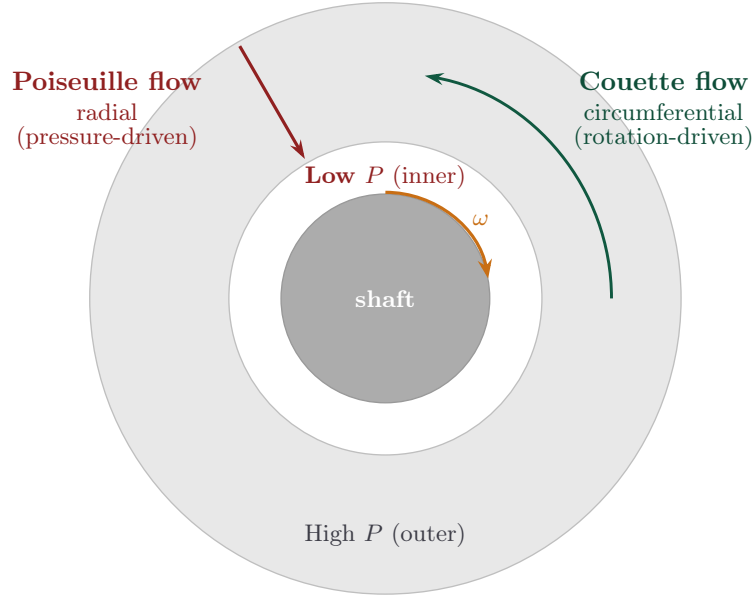


Figure 2: Top view of the seal face ring. Couette flow (green) circulates circumferentially, driven by shaft rotation. Poiseuille flow (red) crosses the annulus radially, driven by the sealed-to-atmosphere pressure difference ΔP . Both flows coexist simultaneously and do not interfere, but they are coupled through the film temperature and thickness h .

How Film Thickness h Is Determined: The Force Balance

The film thickness h is not a parameter that is set during manufacture; it is the *operating solution* that satisfies a force balance on the seal faces. Two sets of forces act axially (perpendicular to the face plane):

Closing forces (pushing faces together):

- Spring load F_s — fixed mechanical force from the closing element.
- Hydraulic closing force: sealed pressure P_i acts on the annular back face of the rotating ring. For a balanced seal with balance ratio B , the net hydraulic closing force per unit face area is approximately $B \Delta P$.

Opening forces (pushing faces apart):

- The average pressure in the gap. For a purely hydrostatic (flat face, no rotation) case with a linear pressure profile, the average is $(P_i + P_o)/2$, giving a net opening force coefficient of approximately $0.5 \Delta P$ per unit area.
- Hydrodynamic lift: additional opening force generated by surface motion and thermal deformation (Section 5).

Equilibrium ($F_{\text{close}} = F_{\text{open}}$) determines h . Because hydrodynamic lift rises sharply as h decreases (hydrodynamic pressure $\propto h^{-2}$ for a wedge geometry, from Reynolds' equation), this equilibrium is *self-stabilising*: if external forces push the faces together, lift increases and pushes them back; if faces drift apart, lift drops and springs close them. This restoring behaviour is the foundation of seal stability.

Typical operating film thickness in water-service contacting seals: $h \approx 0.5\text{--}2\ \mu\text{m}$ (Lebeck, 1991, Chapters 5–7).

Typical leakage: $Q \sim 0.1\text{--}10\ \text{ml/h}$ — “near-zero” in practice.

How Hydrodynamic Lift Is Generated: The Wedge Mechanism

A crucial subtlety must be stated clearly: **two perfectly flat, parallel surfaces sliding past each other generate no net lift**. If the film thickness h is uniform and constant around the face, the Couette term $U \partial h / \partial \theta$ in Eq. (1) is zero and no pressure is generated by the sliding motion. The symmetric pressure rise and fall in a diverging and converging region would cancel exactly under the Sommerfeld boundary condition.

Lift requires a *geometrical asymmetry* in the gap. In real mechanical seals this asymmetry arises from three documented mechanisms:

Thermoelastic Coning (Dominant Mechanism in Wet Contacting Seals)

Frictional heat (power per unit area = $\tau U = \mu \omega^2 r^2 / h$) is concentrated at the inner radius where rubbing velocity is highest. The seal ring heats non-uniformly: the face is hotter than the back, and the inner diameter (ID) is hotter than the outer diameter (OD). Differential thermal expansion across the ring cross-section causes the face to *cone* — rotate about the ring's centroid so that the inner face edge opens or closes relative to the mating face (Figure 4).

For an *externally pressurised* seal (high pressure on the OD), thermoelastic coning typically opens the OD gap and closes the ID, producing a radially *converging* channel from OD to ID. The radial Poiseuille flow must then pass through a narrowing funnel; the minimum film thickness $h_{\min}$ at the ID outlet becomes the controlling restriction. Even if the mean film thickness grows, $h_{\min}$ — and therefore $Q_{\text{leak}} \propto h_{\min}^3$ — can remain small or decrease.

This is the primary mechanism by which real wet seals resist the h^3 leakage increase that simple theory would predict as operating speed rises.

Cavitation Pressure Truncation (JFO Mechanism)

Where the face has circumferential waviness (typically 2–4 waves per revolution, amplitude $\sim 0.1\text{--}1\ \mu\text{m}$), the rotating face alternately converges and diverges relative to the stationary face. In the converging zone, Couette flow generates a positive pressure spike; in the diverging zone, it should generate a corresponding negative trough.

However, a liquid cannot sustain significant tensile stress: when the local pressure falls below the saturated vapour pressure P_{vap} , the film ruptures and *cavitation* occurs. The negative pressure trough is physically truncated at P_{vap} (Jakobsson and Floberg, 1957; Olsson, 1965 — the JFO boundary condition). As a result, the positive and negative pressure contributions no longer cancel: a net positive opening force — hydrodynamic lift — persists. This mechanism is the reason why even a lightly wavy seal face can sustain a fluid film at design speed.

Figure 3 illustrates the pressure truncation that produces net lift.

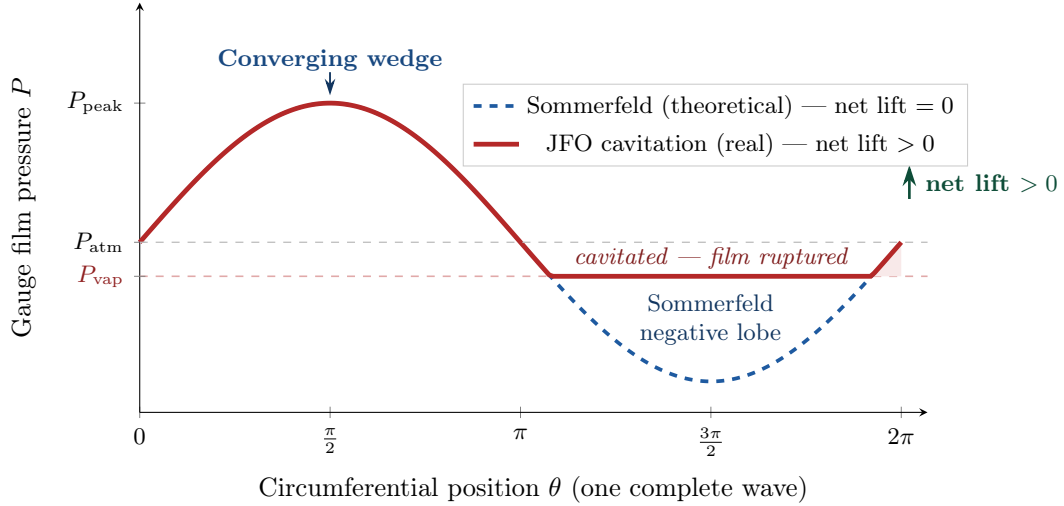


Figure 3: Pressure distribution around one circumferential wave. Under the Sommerfeld condition (dashed, blue) the positive and negative lobes are equal and net lift is zero. With JFO cavitation (solid, red) the negative lobe is truncated at P_{vap} ; the net area under the curve is positive, producing hydrodynamic lift. (After Jakobsson and Floberg, 1957; Olsson, 1965.)

Hydraulic Face Deformation

Sealed pressure loading on the back of the seal ring causes elastic bending. A well-designed balanced seal is shaped so this pressure-induced deformation produces a small, favourable face taper — converging in the direction that adds opening force. This is a hydrostatic effect (present even at zero speed) rather than hydrodynamic, and it is calculated using finite-element methods as part of the seal design process (Lebeck, 1991, Chapter 9).

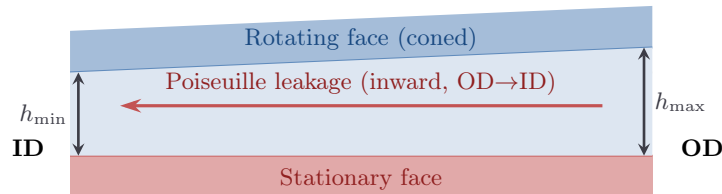


Figure 4: Thermoelastic coning: frictional heat from Couette shear causes the rotating ring to cone so that the OD gap (h_{max}) is larger than the ID gap (h_{min}). For an externally pressurised seal, the Poiseuille leakage flows inward and must pass through the narrowest section h_{min} at the ID. Even if the mean film thickness increases with speed, h_{min} can remain small, limiting leakage. (Based on Lebeck, 1991.)

The h^3 Paradox: Why More Speed Does Not Mean More Leakage

From Eq. (5), $Q \propto h^3$. Higher speed tends to increase hydrodynamic lift and therefore h . Naively this implies higher speed $\Rightarrow$ larger $h \Rightarrow$ more leakage. Yet operators consistently observe the opposite: leakage stabilises or falls as the machine reaches design speed.

Table 1 summarises the four coupled mechanisms that resolve this paradox. They do not violate the h^3 law — they change what h *actually is* at the controlling restriction.

Table 1: Four mechanisms that suppress leakage at operating speed despite the $Q \propto h^3$ scaling law. Each mechanism modifies either the effective h or the effective ΔP seen by Eq. (5).

Mechanism	Physical origin	Effect on leakage	Key reference
1. TEHD Coning (Thermo-Elastohydrodynamic)	Couette shear heat causes non-uniform thermal expansion. Face cones so that the minimum gap $h_{\min}$ (at the ID exit) restricts flow even as mean h grows.	$Q \propto h_{\min}^3$ where $h_{\min}$ is controlled by the coning angle, not the mean gap. Can keep Q near-constant or decreasing despite rising mean h .	Lebeck (1991), Ch. 7–9; Mayer (1977).
2. JFO Cavitation (Jakobsson-Floberg-Olsson)	In the diverging wedge, pressure falls below vapour pressure and the film ruptures. The negative pressure trough is physically truncated, producing net hydrodynamic lift and a stable film.	Enables the seal to run on a well-defined film rather than in boundary contact. Without this mechanism, no stable non-contacting film would exist in wavy-face seals.	Jakobsson & Floberg (1957); Olsson (1965); Hamrock <i>et al.</i> (2004).
3. Centrifugal Inertia	Fluid rotating in the gap acquires centrifugal momentum. The resulting outward body force opposes the inward Poiseuille-driven pressure gradient ($\Delta P_{\text{cf}} = \frac{3}{20}\rho\omega^2(r_o^2 - r_i^2)$).	Reduces effective driving pressure: $\Delta P_{\text{eff}} = \Delta P - \Delta P_{\text{cf}}$. At high speed in an externally pressurised seal, the centrifugal term can approach ΔP , choking inward leakage.	Salant (1993); Buck (1999); Müller & Nau (1998).
4. Vaporisation / Mass-Flow Choke	Intense Couette shear heat raises local film temperature. As pressure drops along the radial leak path, the fluid crosses the saturation curve and flashes to vapour (specific volume $\sim 1600\times$ liquid).	The physical volume constraint of the gap induces a mass-flow choke. Vapour phase “fills” the gap without transmitting net liquid mass — observed as near-zero visible leakage. Not a seal failure mode <i>unless</i> vaporisation is sustained (see Section 8).	Lebeck (1991), Ch. 10; Mayer (1977).

These four mechanisms are not competing explanations of the same phenomenon; they act simultaneously and reinforce each other. Their relative importance depends on seal size, speed, pressure, and fluid properties. TEHD coning dominates in most industrial liquid-service seals; centrifugal inertia becomes significant at shaft speeds above ~ 3000 r/min for large-diameter seals (Müller and Nau, 1998).

The Stribeck Curve: Three Regimes of a Seal’s Life

Tribologists describe lubricated contact using the **Stribeck curve** (Stribeck, 1902): a plot of friction coefficient (and, for seals, leakage) against the dimensionless group $\mu\omega r/P$, the Hersey number. Three regimes are relevant to seals:

1. **Boundary lubrication** (low speed / high load): asperities carry most of the load; fluid film is patchy. Friction is high, wear is active, leakage is through asperity micro-channels. This is the condition at startup and shutdown.
2. **Mixed lubrication**: partial film forms; load shared between asperities and fluid. Friction falls rapidly with speed. Most well-designed contacting wet seals operate at the boundary-to-mixed transition at design speed — they do not run fully non-contacting.
3. **Full-film hydrodynamic**: faces fully separated by a continuous film. Wear ceases, but film is thicker and $Q \propto h^3$ leakage is higher. This is the intended regime for *non-contacting* seals (gas seals, spiral-groove liquid seals).

The practical implication is that a contacting wet seal is designed to live in the mixed regime: it allows microscopic asperity contact (and thereby keeps h very small) without generating damaging wear rates. The four paradox-resolving mechanisms of Section 5 are what prevent the film from becoming so thin that catastrophic dry-running results, or so thick that leakage becomes unacceptable.

Seal Failure Modes

Understanding how seals fail is as important as understanding how they work. The common failure modes all trace back to disruption of the thin-film equilibrium.

Dry running. If the film evaporates, boils, or is never established (startup without flush, improper installation), the faces contact dry. Frictional power $P_f = \tau U \cdot A_{\text{face}}$ spikes; SiC faces develop *heat checks* (radial thermal cracks from rapid localised heating and quenching); carbon faces blister or graphitise. Failure can occur in seconds to minutes. The audible signature is a high-pitched squeal.

Flash vaporisation (sustained). If sustained vaporisation occurs throughout the film (rather than momentary mass-flow choke), the vapour film provides no lubrication. The result is thermally driven dry running. This distinguishes “beneficial” momentary flashing (mechanism 4 in Table 1) from destructive sustained dry-vapour running.

Excess axial shaft movement. The spring or bellows has a finite axial travel budget. If the shaft moves axially (due to hydraulic thrust, bearing clearance, or transient events) beyond this budget:

- Shaft moves toward the stationary face $\Rightarrow$ faces over-compressed $\Rightarrow$ film squeezed out $\Rightarrow$ dry running and heat.
- Shaft moves away $\Rightarrow$ spring cannot extend far enough $\Rightarrow$ faces separate $\Rightarrow h$ grows $\Rightarrow h^3$ leakage jump.

This is why axial movement tolerance is a *critical* seal selection parameter, distinct from and equally important as pressure, speed, and temperature ratings.

Secondary seal failure. O-ring hang-up (adhesion to a rough shaft), extrusion at high pressure, chemical swell, and heat hardening can prevent the dynamic secondary seal from sliding, locking the rotating face in a fixed axial position and eliminating its ability to track shaft float.

Misalignment / TIR. Shaft radial runout (Total Indicated Runout, TIR) imposes a cyclic tilting of the rotating face relative to the stationary face. If TIR is excessive, the faces cannot maintain full contact around the circumference; one sector loses the film and leaks or heats while the opposite sector is over-loaded. API 682 limits TIR to ≤ 0.05 mm for standard installations.

Important distinction. *Radial TIR* (shaft wobble, measured with a dial indicator against the rotating shaft OD) and *axial movement tolerance* (lengthwise shuttle of the shaft) are entirely different quantities on different axes. They appear on separate lines of a seal datasheet and have separate physical effects. Confusing them — for example, assuming a published TIR figure addresses the axial movement question — is a common and consequential error in seal selection.

Worked Example: Surface Speed and Leakage Estimate

Given: shaft diameter $d = 160$ mm, rotational speed $N = 1000$ r/min, sealed pressure $\Delta P = 5$ bar = 500 kPa, water viscosity $\mu = 8.9 \times 10^{-4}$ Pa s, seal face width $b = r_o - r_i = 15$ mm, and estimated film thickness $h = 1$ μ m.

Surface speed:

$$U = \pi d N / 60 = \pi \times 0.160 \times 1000 / 60 \approx 8.4 \text{ m/s}$$

Shear stress and heat flux at the face:

$$\tau = \mu U / h = (8.9 \times 10^{-4} \times 8.4) / (1 \times 10^{-6}) \approx 7480 \text{ Pa} \quad ; \quad q = \tau U \approx 62,800 \text{ W/m}^2 = 6.3 \text{ W/cm}^2$$

Radial Poiseuille leakage (Eq. (5)), with $r_i = 80$ mm, $r_o = 95$ mm:

$$Q = \frac{\pi \times (1 \times 10^{-6})^3 \times 500,000}{6 \times 8.9 \times 10^{-4} \times \ln(95/80)} \approx 2.1 \times 10^{-10} \text{ m}^3/\text{s} \approx 0.76 \text{ ml/h}$$

This sub-millilitre-per-hour figure is near-zero in practice and confirms why well-designed seals are considered essentially non-leaking. Notice also the sensitivity: if h doubles to 2 μ m, leakage rises by $2^3 = 8$ times to ~ 6 ml/h — still small, but eight times larger. This is the h^3 sensitivity in action.

Why This All Matters: Selection, Assessment, and Consequence

The mechanical face seal is one of the few machine components where micrometre-scale film physics has direct operational consequence at the plant level. Selecting the wrong seal — or substituting a seal without engineering assessment — can displace a machine's operating point from the stable mixed-lubrication regime into dry running (immediate noise and damage) or sustained leakage (process loss, environmental incident, plant shutdown).

The selection parameters that must all be verified in combination are:

- **Pressure rating:** maximum allowable sealed ΔP .
- **Surface speed (PV limit):** maximum allowable face velocity $U = \pi d N$, and pressure-velocity product PV .

- **Temperature:** operating range of the process fluid, determining viscosity, vapour pressure, and elastomer compatibility.
- **Radial TIR:** maximum allowable shaft runout (a radial parameter).
- **Axial movement tolerance:** maximum allowable shaft axial float (an *axial* parameter, orthogonal to TIR and equally critical).
- **Flush plan** (API 682): how the seal chamber is cooled, flushed, and protected from contamination.

A seal that appears adequate on every parameter except axial movement will fail on that axis alone. The film physics is unforgiving: once the axial budget is exceeded, the force balance collapses, and the machine announces the fact with a squeal and a leak — both at once if the shaft oscillates through the spring’s travel limit in each direction.

Conclusion

The mechanical face seal is a masterpiece of applied tribology. A film a single micrometre thick, governed by the Reynolds lubrication equation, simultaneously seals against pressurised process fluid, lubricates the face interface, and carries away frictional heat — as long as all the conditions for film stability are met. The apparent paradox that leakage does not grow with speed is resolved by four coupled high-order mechanisms: thermoelastic coning, JFO cavitation, centrifugal inertia, and vaporisation-induced mass-flow choke. These mechanisms do not override the $Q \propto h^3$ law; they regulate h and the effective driving pressure so that the law produces a small, stable leakage rate at design conditions.

Failure, when it comes, is almost always a failure of equilibrium: the film is either destroyed (dry running, heat, damage) or excessively opened (axial movement beyond the spring budget, misalignment, or wrong seal selection). Understanding the physics at the scale of a micrometre is what allows engineers to avoid those failures at the scale of a plant.

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